Monetary Policy Shocks and Macroeconomic Dynamics in Switzerland

A Replication and Extension of Stock and Watson (2001)

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Country: Switzerland

Frequency: Quarterly

Sample: 1999Q3 – 2025Q2

1 Introduction

Stock and Watson (2001) study the empirical effects of monetary policy shocks using vector autoregressions (VARs) and alternative identification strategies. Their contribution emphasizes the robustness of impulse responses across models and highlights the limits of strong structural assumptions.

This paper closely replicates the empirical framework of Stock and Watson (2001) using Swiss data. Switzerland provides an interesting case study due to its long-standing low inflation environment, credible monetary policy, and relatively stable macroeconomic structure. The main objective is to assess whether the qualitative findings of Stock and Watson (2001), originally obtained for the United States, carry over to a small open economy with an independent central bank.

The first part of the paper follows the original analysis as closely as possible. The second part extends the framework to include alternative identification schemes, local projections, and non-stationarity considerations, as required by the project assignment.

2 Data and Variable Construction

The empirical analysis is based on quarterly Swiss data from 1999Q3 to 2025Q2. The three variables included in the baseline VAR are inflation, unemployment, and the short-term interest rate.

Inflation is measured as the annualized quarterly growth rate of the GDP deflator. The unemployment rate is expressed in percentage points and captures labor market conditions. The interest rate corresponds to the three-month money market rate and serves as the monetary policy instrument.

This choice of variables mirrors the baseline three-variable VAR in Stock and Watson (2001). Figure 1 plots the three series. Inflation displays moderate volatility, unemployment fluctuates around a stable mean, and interest rates exhibit long periods near the zero lower bound, reflecting Switzerland's monetary regime.

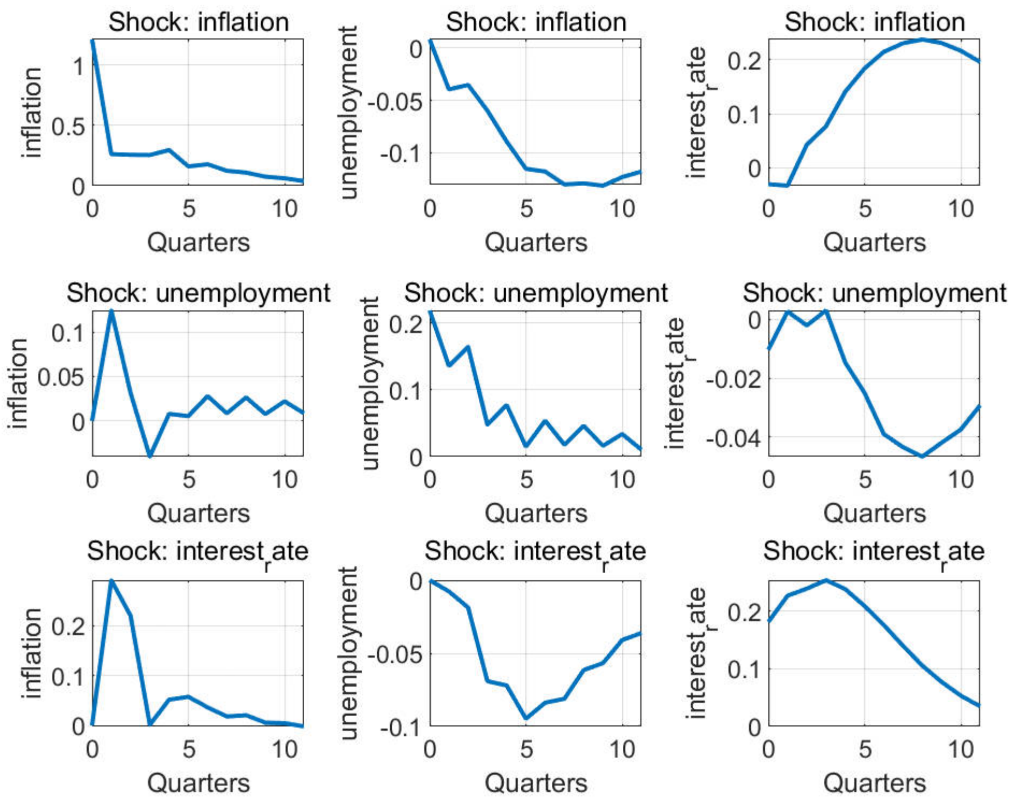

Figure 1: Macroeconomic variables in Switzerland: inflation, unemployment, and the short-term interest rate.

3 VAR Specification and Lag Length Selection

Let y_t denote the 3×1 vector containing inflation, unemployment, and the interest rate. The reduced-form VAR of order p is given by

$$y_t = c + A_1 y_{t-1} + \cdots + A_p y_{t-p} + u_t,$$

where u_t is a vector of reduced-form innovations with covariance matrix Σ_u.

Following Stock and Watson (2001), the lag length is selected using information criteria. Although the Bayesian Information Criterion favors a more parsimonious specification, a VAR(4) is adopted as the baseline to maintain comparability with the original study and to allow for richer dynamics.

Granger causality tests indicate that inflation helps predict the interest rate and unemployment, while feedback from unemployment to inflation is weak. These results broadly resemble the predictive relationships documented in the original paper.

4 Summary Statistics and Predictability

Table 1 reports summary statistics for the three macroeconomic variables. Inflation exhibits moderate volatility, while unemployment and interest rates display greater persistence. These features are broadly consistent with the descriptive evidence reported in Stock and Watson (2001) for the United States.

Table 1: Summary statistics

	Inflation	Unemployment	Interest rate
Mean	0.51	4.21	0.46
Std. dev.	1.44	0.72	1.16

Table 2 reports the information criteria used to select the VAR lag length. While the Bayesian Information Criterion favors a more parsimonious specification, a VAR(4) is retained as the benchmark, following Stock and Watson (2001).

Table 2: Lag length selection

Lag	AIC	BIC
1	416.5	448.2
2	347.4	403.0
3	334.8	414.1
4	**322.2**	425.3

Following Stock and Watson (2001), out-of-sample forecast performance is evaluated by comparing a random walk (RW), an autoregressive model (AR), and the VAR.

Table 3: Pseudo out-of-sample RMSE

	RW	AR(4)	VAR(4)
Inflation (8Q)	1.35	1.20	**1.17**
Unemployment (8Q)	**0.39**	0.44	0.57
Interest rate (8Q)	**0.29**	0.54	0.92

5 Impulse Responses: Replication of Stock and Watson (2001)

This section closely follows the impulse response analysis in Stock and Watson (2001). Monetary policy shocks are identified using a recursive (Cholesky) ordering in which the interest rate is ordered last. This identification assumes that monetary policy reacts contemporaneously to macroeconomic conditions, while inflation and unemployment respond with a lag.

Figure 2 reports the impulse responses of inflation, unemployment, and the interest rate to one-standard-deviation shocks. An inflation shock raises inflation on impact and leads to a temporary decline in unemployment. An unemployment shock increases unemployment persistently and induces a gradual response of inflation. A monetary policy shock leads to a persistent increase in the interest rate, a decline in inflation, and a rise in unemployment.

The qualitative responses are consistent with standard monetary transmission mechanisms and broadly resemble those reported by Stock and Watson (2001) for the United States, although the magnitudes are smaller.

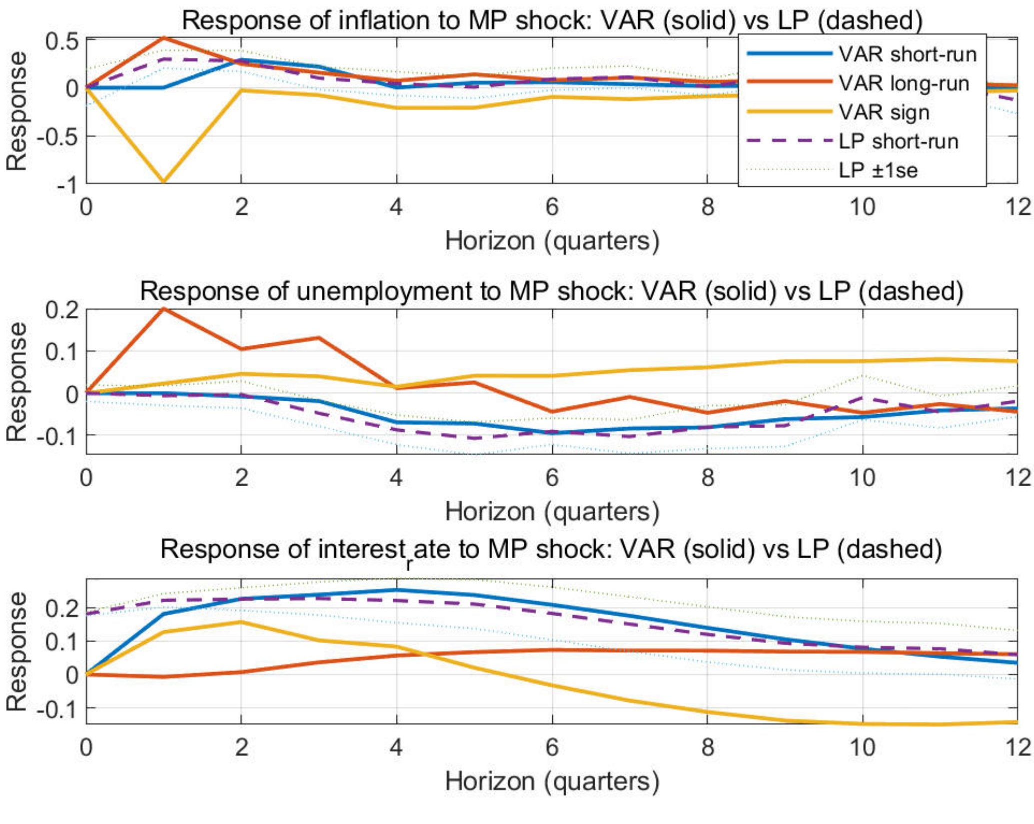

Figure 2: Impulse responses from the baseline VAR(4). This figure corresponds to Figure 3 in Stock and Watson (2001).

6 Structural VAR and Monetary Policy Rule Interpretation

Stock and Watson (2001) emphasize that monetary policy shocks should be interpreted as deviations from a systematic policy rule, rather than arbitrary movements in interest rates. In this framework, the short-term interest rate responds endogenously to inflation and real activity, consistent with a Taylor-type rule.

Formally, the monetary policy equation can be written as

$$i_t = \phi_\pi \pi_t + \phi_u u_t + \varepsilon_t^{MP},$$

where ε_t^{MP} represents an exogenous monetary policy shock.

The recursive identification adopted in the baseline VAR is consistent with this interpretation. By ordering the interest rate last, the model allows the central bank to respond contemporaneously to inflation and unemployment, while private-sector variables respond to policy innovations only with a lag.

Under this interpretation, the identified monetary policy shock captures unexpected deviations from the systematic component of policy, rather than predictable responses to macroeconomic conditions. This perspective is central to the interpretation of impulse responses in Stock and Watson (2001).

7 Forecast Error Variance Decompositions

To further assess the importance of monetary policy shocks, forecast error variance decompositions (FEVDs) are computed. The FEVD measures the fraction of the h-step-ahead forecast error variance of each variable attributable to each structural shock.

Figure 3 presents the FEVDs for inflation, unemployment, and the interest rate. At short horizons, inflation variability is primarily driven by its own shocks, while monetary policy shocks play a limited role. For unemployment, non-monetary shocks dominate at all horizons. In contrast, the interest rate is largely explained by its own innovations, particularly at short horizons.

These findings closely mirror Figure 4 in Stock and Watson (2001) and suggest that, while monetary policy shocks have clear dynamic effects, they account for only a modest share of overall macroeconomic volatility.

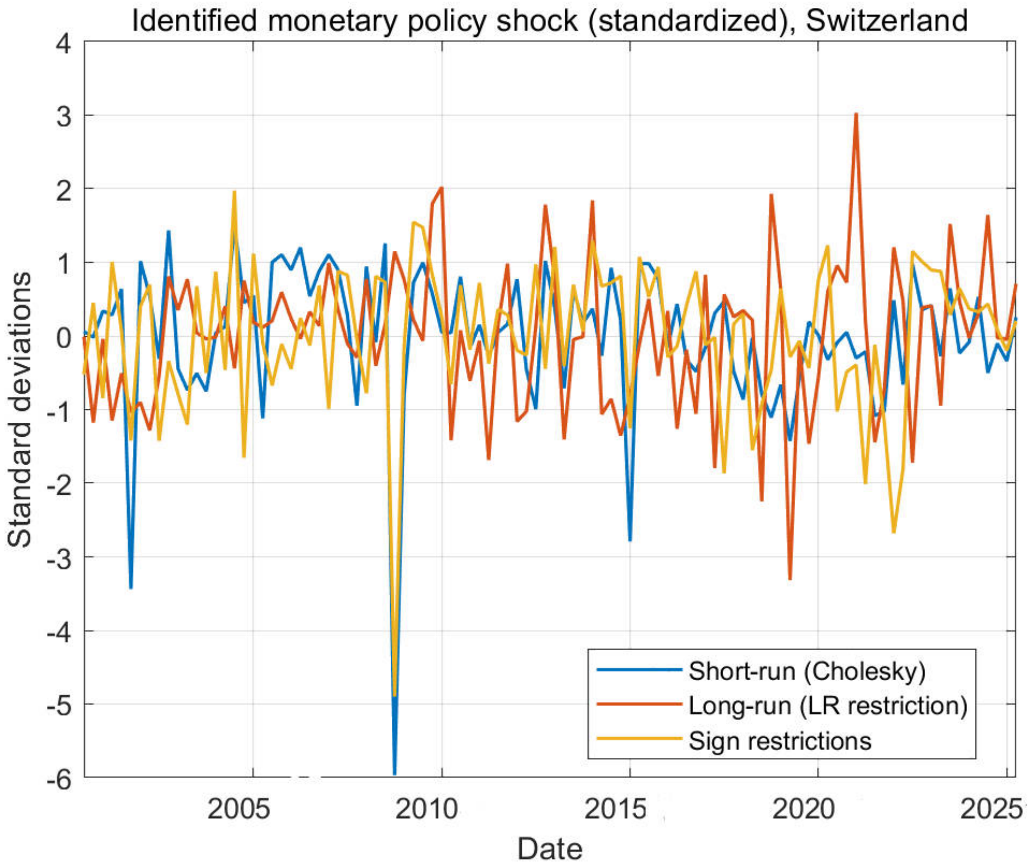

Figure 3: Forecast error variance decompositions from the baseline VAR(4). This figure corresponds to Figure 4 in Stock and Watson (2001).

The impulse response patterns admit a clear economic interpretation. A contractionary monetary policy shock leads to an immediate and persistent increase in the short-term interest rate. Inflation responds with a delay and gradually declines, reflecting nominal rigidities and the slow pass-through of policy to prices. Unemployment rises over subsequent quarters, indicating that tighter monetary conditions dampen real economic activity.

This dynamic adjustment is consistent with standard New Keynesian transmission mechanisms. Monetary tightening raises borrowing costs and suppresses aggregate demand, which in turn reduces price pressures and increases slack in the labor market. Compared to the United States, the responses in Switzerland are smaller in magnitude, which may reflect the credibility of monetary policy, lower inflation persistence, and the openness of the Swiss economy.

The forecast error variance decompositions reinforce this interpretation. At short horizons, inflation and unemployment fluctuations are dominated by their own shocks, while monetary policy shocks account for only a limited share of total variance. This finding echoes one of the central messages of Stock and Watson (2001): although monetary policy shocks have clear dynamic effects, they explain only a modest fraction of macroeconomic volatility.

For the interest rate, own shocks dominate at all horizons, consistent with the interpretation of monetary policy as largely systematic, with relatively small exogenous innovations.

8 Extensions: Alternative Identification Schemes

This section extends the baseline analysis to consider alternative identification strategies. In addition to the recursive scheme, two further approaches are implemented: a long-run restriction imposing monetary neutrality on unemployment, and a sign-restriction approach consistent with contractionary monetary policy shocks.

Figure 4 plots the standardized monetary policy shocks obtained from the three schemes. While the identified shocks share some common features, their correlation is far from perfect, indicating that identification choices matter quantitatively.

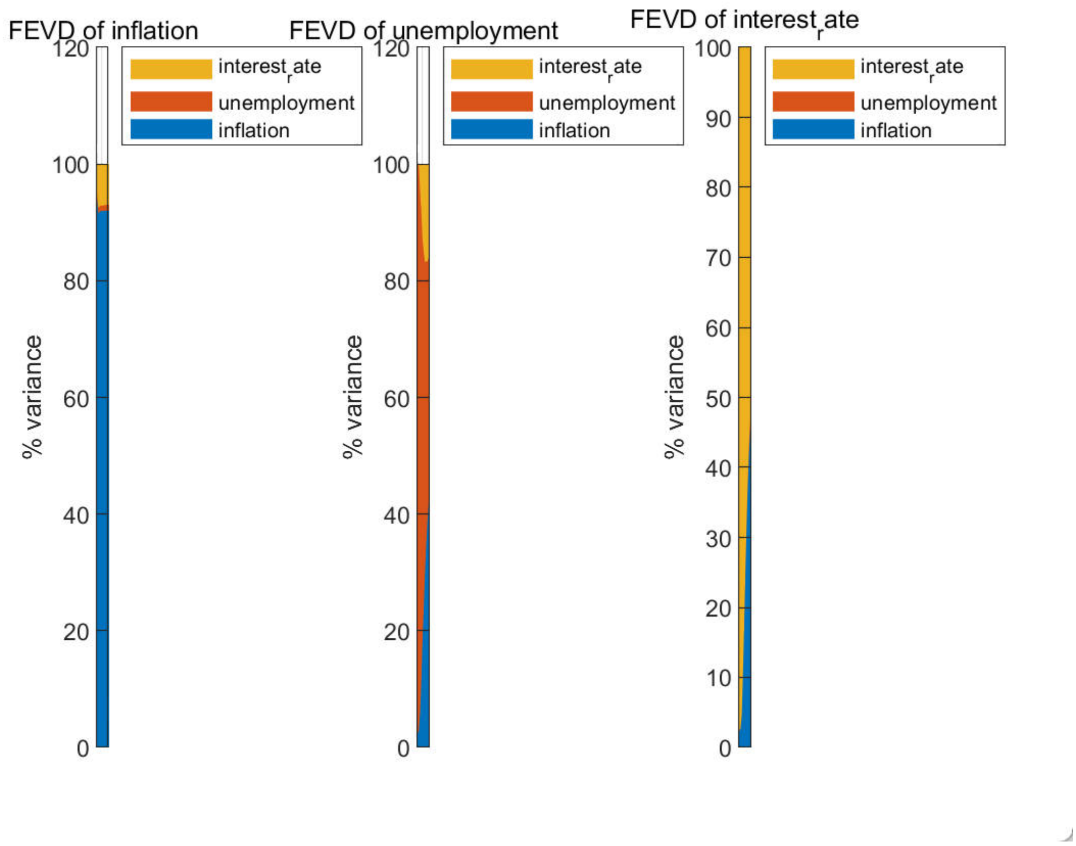

Figure 4: Identified monetary policy shocks under alternative identification schemes.

9 VAR versus Local Projections

Following recent empirical practice, impulse responses obtained from the VAR are compared with those estimated using local projections. Local projections estimate each horizon separately and are therefore less sensitive to VAR misspecification.

The comparison reveals that VAR-based and LP-based impulse responses are generally similar. In most cases, the VAR responses lie within the LP confidence bands, suggesting that the dynamic effects are robust to the estimation method.

10 Non-Stationarity and Cointegration

Unit root tests indicate that inflation is stationary, whereas unemployment and the interest rate are close to unit-root processes. Johansen cointegration tests provide strong evidence of a cointegrating relationship between unemployment and the interest rate.

These results suggest that a Vector Error Correction Model could be considered. However, given the focus on short- to medium-run dynamics and the mixed integration properties of the variables, the VAR in levels remains a reasonable approximation, consistent with the approach in Stock and Watson (2001).

11 Conclusion

This paper replicates and extends the empirical analysis of Stock and Watson (2001) using Swiss data. The main qualitative findings of the original study carry over to Switzerland: monetary policy shocks have economically meaningful effects on inflation and unemployment but explain only a limited share of overall macroeconomic volatility.

Extensions using alternative identification schemes and local projections confirm the robustness of the main conclusions, while also highlighting the importance of identification assumptions. Overall, the evidence suggests that Swiss monetary policy has been effective in influencing macroeconomic outcomes, though its quantitative impact depends on modeling choices.

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